NICOLE LEWIS

nicole.lewis@rice.edu

RESEARCH INTERESTS

Experimental nuclear and particle physics at hadron colliders (LHC, RHIC) focusing on fundamental aspects of quantum chromodynamics (QCD) and its emergent properties in small systems; ultra-peripheral collisions; development of fast-timing silicon detectors and novel Level-1 triggers for the future high-luminosity era of the LHC and EIC physics

EDUCATION

Ph.D. in Physics – August 2015 to August 2020
University of Michigan, Ann Arbor
Thesis: "Transverse Single-Spin Asymmetries of Midrapidity Direct Photons, Neutral Pions, and Eta Mesons in 200 GeV Polarized Proton-Proton Collisions at PHENIX" arXiv:2008.04283 [hep-ex]
Advisor: Professor Christine Aidala

B.A. in Physics, Minor in Forestry and Natural Resources – August 2011 to May 2015
University of California, Berkeley
Advisors: Professor Michael DeWeese, Professor Joel Fajans

RESEARCH POSITIONS

Postdoctoral Researcher at Rice University
Rice Academy Postdoctoral Fellow
CMS – *July 2023 to Present*

- Observed a long-range near-side peak in high-multiplicity photonuclear collisions. This two-particle correlations analysis was performed on PbPb data and was the first clear indication of high-energy density QCD dynamics happening in photonuclear collisions.
- Measurement of identical-particle femtoscopic correlations in high-multiplicity photonuclear collisions in order to probe the spatial dimensions of these photonuclear collisions.
- Measurement of the J/ψ and $\psi(2s)$ cross sections in coherent ultraperipheral PbPb collisions, which are sensitive to the internal structure of these nuclei
- Managed project on the Endcap Timing Layer (ETL) detector for the CMS upgrade, designed the high-voltage map, maintained a test stand for developing front-end electronics, and took data to test those electronics at a test beam facility.
- Led initiative to push the Level-1 tracking trigger for the CMS upgrade to lower momentum tracks. This trigger would become one of the main triggers for recording heavy-ion collision data at CMS.

Research Associate at Brookhaven National Laboratory
STAR – *September 2020 to June 2023*

- Implemented initial analysis of baryon stopping in inclusive photonuclear collisions in AuAu data. This measurement was found to be sensitive to the baryon number carrier and inconsistent with the conventional model of the valence quarks carrying baryon number and has been published as ***Science* 393, 727 (2026)**

- Contributed to the commissioning and calibration of the Small-Strip Thin Gap Chamber (sTGC) for the STAR forward upgrade.

Graduate Student Research Assistant at the University of Michigan, Ann Arbor
PHENIX, sPHENIX – *May 2016 to August 2020*

- Wrote a thesis on analyses of spin-momentum correlations in transversely polarized proton-proton collisions recorded by PHENIX using direct photons, neutral pions, and eta mesons.
- Developed and maintained a quality assurance tester for silicon photomultipliers (SiPMs) for the sPHENIX calorimeters. Managed the Michigan team by coordinating a testing schedule for team members.

MENTORSHIP EXPERIENCE

Rice University

- September 2026 to Present – Hajime Kioi, Undergraduate Student, *Rice University*
- August 2026 to Present – Aryaa Dattamunsi, Visiting Graduate Student, *Indian Institute of Technology Madras*
- January 2025 to Present – Isabela Maietto Silverio, Visiting Graduate Student, *SPRACE-UNESP*
- August 2024 to Present – Rachel Kovach-Fuentes, Graduate Student, *Rice University*
- October 2023 to Present – Daniela Hikari Yano, Graduate Student, *Rice University*
- June 2026 to August 2026 – Evan Yang, High School Student
- June 2026 to August 2026 – Adittya Raghuraj, Undergraduate Student, *University of Texas at Austin*
- August 2024 to May 2025 - Wren Kawamura, Undergraduate Student, *Rice University*

Brookhaven National Laboratory

- June 2021 to June 2022 – Isabel Xu, High School Student

University of Michigan

- Feb 2019 to March 2023 – Dillon Fitzgerald, Graduate Student, *University of Michigan*
- August 2018 to May 2019 - Desmond Shangase, Graduate Student, *University of Michigan*
- August 2017 to May 2018 - Jordan Roth, Graduate Student, *University of Michigan*

TEACHING EXPERIENCE

Graduate Student Instructor, Fall 2017
Physics 405 – Intermediate Electricity and Magnetism
Ran two discussion sections for the intermediate electrodynamics course. This was the first time this course offered discussion sections, so I developed the course structure and content myself, including worksheets and quizzes.

Graduate Student Instructor, Winter 2017
Physics 136 – Life Sciences Mechanics Laboratory

Graduate Student Instructor, Fall 2016
Physics 141 – Elementary Laboratory Mechanics

INVITED CONFERENCE AND WORKSHOP PRESENTATIONS

APS Division of Nuclear Physics, Fall Meeting (DNP 2025), Chicago, Illinois, October 2025
Experimental Overview on Baryon Transport in High-Energy Nuclear Collisions

The 4th International Workshop on QCD Collectivity at the Smallest Scales, Qingdao, China, June 2024, *Experimental results of collectivity in UPC, DIS, and Photoproduction*

The 1st International Workshop on TeV Muon-Ion Colliders, Houston, Texas, December 2023, *Nucleon Spin Physics with Polarized Beams: Past, Present and Future*

14th Conference on the Intersections of Particle and Nuclear Physics (CIPANP 2022), Lake Buena Vista, Florida, September 2022, *New Results from the RHIC-Spin Program*

Annual RHIC and AGS Users Meeting, BNL, June 2022, *UPC Measurements at STAR*

APS Division of Nuclear Physics, Fall Meeting (DNP 2021), Online Conference, October 2021
New Results from the RHIC Spin Program

SEMINARS AND COLLOQUIA

Physics Department Colloquium at Old Dominion University, January 30, 2026**,** *Probing the Internal Structure of the Proton through Transverse Single-Spin Asymmetries and Photonuclear Collisions*

Physics Department Colloquium at Creighton University, November 20, 2025, *Probing the Strong Force Through Photonuclear Collisions*

Physics Division Seminar at Argonne National Laboratory, November 19, 2024, *Probing the Internal Structure of the Proton through Transverse Single-Spin Asymmetries and Baryon Stopping*

Nuclear Seminar at the University of Kentucky, October 31, 2024, *Search for baryon number carrier in photonuclear processes and heavy-ion collisions at STAR*

Nuclear Particle Astrophysics (NPA) Seminar at Yale University, May 2, 2024, *Search for baryon junctions in photonuclear processes and heavy-ion collisions at STAR*

Center for Frontiers in Nuclear Physics Seminar, May 25, 2023, *Search for baryon junctions in photonuclear processes and heavy-ion collisions at STAR*

Hadron Ion Tea (HIT) Seminar at Laurence Berkeley National Laboratory, February 9, 2023, *Baryon Stopping in Photonuclear Collisions*

Physics and Astronomy Department NPP Seminar at Rice University, January 31, 2023, *Baryon Stopping in Photonuclear Collisions*

Particle-Astro-Nuclear Seminar at Wayne State University, November 11, 2022, *Baryon Stopping in Photonuclear Collisions*

Nuclear Physics Seminar at Brookhaven National Laboratory, November 30, 2021, *Probing the gluon structure of the nucleon through baryon stopping in photonuclear collisions*

Los Alamos National Lab P-25 Seminar, January 7, 2020, *Transverse Single-Spin Asymmetries of Midrapidity Direct Photons, Neutral Pions, and Eta Mesons at PHENIX*

OTHER CONFERENCE PRESENTATIONS

The second international workshop on the physics of Ultra Peripheral Collisions (UPC 2025), Saariselkä, Finland, June 2025, *Probing QCD Collectivity over the Widest Rapidity Gap in Photonuclear Collisions at CMS*

Initial Stages (IS 2023), Copenhagen, Denmark, June 2023, *STAR Overview* (Plenary talk)

Deep Inelastic Scattering (DIS 2023), East Lansing, MI, March 2023, *Search for baryon junctions in photonuclear processes and heavy-ion collisions at STAR*

APS Division of Nuclear Physics, Fall Meeting (DNP 2022), New Orleans, LA, October 2022, *Identified hadron spectra and baryon stopping in $\gamma + Au$ collisions at STAR*

XXIXth International Conference on Ultra-relativistic Nucleus-Nucleus Collisions (Quark Matter 2022), Hybrid Conference: Krakow, Poland, April 2022, *Identified hadron spectra and baryon stopping in $\gamma +$* Au *collisions at STAR* (Poster Award Flash Talk)

APS Division of Nuclear Physics, Fall Meeting (DNP 2021), Online Conference, October 2021 *Identification of Pions, Kaons, and Protons in Photonuclear Events at STAR*

Deep Inelastic Scattering (DIS 2021), Online Conference, April 2021, *Transverse Single-Spin Asymmetries of Midrapidity Direct Photons and Neutral Mesons at PHENIX*

Annual RHIC and AGS Users Meeting, BNL, June 2019, *Transverse Spin in PHENIX*

Deep Inelastic Scattering (DIS 2019), Torino, Italy, April 2019, *Transverse Single-Spin Asymmetries of Midrapidity Eta Mesons at PHENIX*

Rencontres de Moriond: QCD and High Energy Interactions (Moriond 2019), La Thuile, Italy, March 2019, *Studying Parton Dynamics Via Single-Spin Asymmetries and Two-Particle Correlations at PHENIX*

International Spin Symposium (Spin 2018), Ferrara, Italy, September 2018, *Transverse Single-Spin Asymmetries of Midrapidity Eta Mesons at PHENIX*

APS Division of Nuclear Physics, Fall Meeting (DNP 2017), Pittsburgh, PA, October 2017, *Transverse Single-Spin Asymmetries for Direct Photon and Neutral Pion Production in Midrapidity at PHENIX*

PUBLIC TALKS

Nerd Nite Ann Arbor #61, January 2019, *Unpacking the Proton* https://aadl.org/aadl_events_20190117-nerd_night_proton_power

Physics Graduate Student Symposium, August 2018, *The Structure of the Proton Through Transverse Single Spin Asymmetries*

SERVICE WORK AND OTHER EXPERIENCE

Co-organizer for 1st Workshop on Baryon Dynamics from RHIC to EIC, January 2024 - Held at the Center for Frontiers in Nuclear Science at Stony Brook University

STAR tour guide, July 2022 to February 2023

Co-organizer for the STAR Collaboration meeting, September 2021 and September 2022

Co-convener for BNL EIC Weekly Meetings at BNL, August 2021 to January 2022

Co-organizer for the Spin Symposium for the RHIC & AGS Annual Users Meeting 2021, June 2021

Graduate Employee Organization Steward for the Physics and Applied Physics Ph.D. Programs at U. of Michigan, January 2019 to May 2020 - Advocated for our departments' needs at union meetings and kept the graduate students in physics and applied physics informed about what was happening in the union through email campaigns and holding department meetings

Participated in RELATE at U. of Michigan, Summer 2018 - Ten-week intensive communication fundamentals workshop focused on teaching STEM graduate students how to communicate their research to a lay audience

U. of Michigan Society for Women in Physics (SWiP), January 2015 to August 2020 - Organized and developed materials for internal department workshops, ran physics demos for surrounding communities, and created the posters that were added to the department halls describing the histories of some of the first women and Black people to earn graduate Physics degrees from Michigan

AWARDS

Rice Academy Fellowship, Rice University, March 2023

Quark Matter 2022 Best Poster Award, one of the 10 posters that were selected out of a total of 500 posters given at Quark Matter that year, April 2022

Baer Fellowship, University of Michigan Physics Department, April 2015

Dean's Honors, University of California, Berkeley, Letters and Sciences Undergraduate Dean, December 2013

PUBLICATION LIST

Publications not with CMS, STAR, or PHENIX:

- "ePIC Early Science Report", D. Abbott *et al.,* arXiv:2608.05377 [nucl-ex] (2026)
- "Search for baryon junctions in photonuclear processes and isobar collisions at RHIC," N. Lewis, *et al., Eur. Phys. J. C* **84,** 560 (2024)
- "Search for Dark Photons in Polarized $\gamma\gamma \rightarrow e^+e^-$ at RHIC," I. Xu, N. Lewis, X. Wang, J. D. Brandenburg, and L. Ruan, arXiv:2211.02132 [hep-ex] (2022)
- "ATHENA detector proposal – a totally hermetic electron nucleus apparatus proposed for IP6 at the Electron-Ion Collider," J. Adam *et al., JINST* **17**, P10019 (2022)

- "Design and Beam Test Results for the 2D Projective sPHENIX Electromagnetic Calorimeter Prototype," C. Aidala *et al., IEEE Trans. Nucl. Sci.* **68**, 173 (2021)
- "Plasma Temperature Measurement with a Silicon Photomultiplier (SiPM)," E. D. Hunter, J. Fajans, N. A. Lewis, A. P. Povilus, C. Sierra, C. So, and D. Zimmer, *Review of Scientific Instruments* **91**, 103502 (2020)

CMS Collaboration Papers:

- "Search for Higgs boson compositeness effects in high-mass dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV and 13.6 TeV," A. Gevorgyan *et al.,* arXiv:2610.04782 [hep-ex] (2026)
- "Search for pair-produced vector-like top quarks decaying into Lorentz-boosted top quarks and scalar bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2610.03489 [hep-ex] (2026)
- "Search for dark matter produced in association with a low-multiplicity jet in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2610.00995 [hep-ex] (2026)
- "Measurement of the triple-differential cross section of $Z +$ jet production in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2609.39653 [hep-ex] (2026)
- "Search for resonant and nonresonant production of pairs of dijet resonances with b jets in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2609.21850 [hep-ex] (2026)
- "Calibration of electromagnetic shower features in the CMS calorimeter with machine-learning techniques," A. Gevorgyan *et al.,* arXiv:2609.15377 [hep-ex] (2026)
- "Search for a light pseudoscalar Higgs boson in final states with boosted muon pairs and tau lepton pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2609.12683 [hep-ex] (2026)
- "Search for the rare Higgs boson decay $H \to Z\gamma$ in proton-proton collisions at $\sqrt{s} = 13$ TeV and 13.6 TeV," A. Gevorgyan *et al.,* arXiv:2609.04402 [hep-ex] (2026)
- "Spin-dependent azimuthal asymmetry of coherent J/ψ photoproduction in hadronic PbPb collisions at $\sqrt{s_{NN}} = 5.36$ TeV," A. Belyaev *et al.,* arXiv:2609.04376 [nucl-ex] (2026)
- "Constraint on the Higgs boson width in the diphoton final state using signal-background interference in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2609.01385 [hep-ex] (2026)
- "Evidence for the dead-cone effect in bottom quark initiated jets produced in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2608.29412 [hep-ex] (2026)
- "Search for scalar leptoquarks produced via muon-quark scattering in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2608.28377 [hep-ex] (2026)
- "Search for top quark flavor-changing neutral currents in multilepton final states using an effective field theory approach in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2608.19547 [hep-ex] (2026)
- "Observation of an excess at the top quark pair production threshold in the single-lepton channel," A. Gevorgyan *et al.,* arXiv:2608.19546 [hep-ex] (2026)
- "Performance of heavy-flavour jet identification in the CMS high-level trigger in proton-proton collisions at $\sqrt{s} = 13.6$ TeV," A. Gevorgyan *et al.,* arXiv:2608.18331 [hep-ex] (2026)
- "Search for anomalous couplings in WW and WZ production with single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2608.18245 [hep-ex] (2026)
- "Search for a resonance in events with four top quarks decaying into two leptons and jets in proton-proton collisions," A. Gevorgyan *et al.,* arXiv:2608.10148 [hep-ex] (2026)

- "Search for resonant production of lepton-enriched semivisible jets in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2608.05323 [hep-ex] (2026)
- "Combination of vector boson scattering measurements in leptonic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2608.01289 [hep-ex] (2026)
- "Search for a top-philic Z' boson decaying into a $t\bar{t}$ pair in a final state with jets and an electron or muon in proton-proton collisions at $\sqrt{s} = 13.6$ TeV," A. Gevorgyan *et al.,* arXiv:2608.01251 [hep-ex] (2026)
- "A measurement of the Higgs boson mass in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2607.28396 [hep-ex] (2026)
- "Measurement of the fragmentation properties of jets containing $\Upsilon(nS)$ mesons in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2607. 25653 [hep-ex] (2026)
- "Off-shell Higgs boson measurements: Yukawa couplings, self-coupling, compositeness, and width," A. Gevorgyan *et al.,* arXiv:2607. 23352 [hep-ex] (2026)
- "Measurement of the $\gamma\gamma \to \tau\tau$ cross section and constraints on the anomalous magnetic moment of the τ lepton in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV," A. Belyaev *et al.,* arXiv:2607.20148 [nucl-ex] (2026)
- "Study of χ_c production in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV," A. Belyaev *et al.,* arXiv:2607.17847 [nucl-ex] (2026)
- "Evidence for sequential $\Upsilon(nS)$ suppression in light ion collisions," A. Belyaev *et al.,* arXiv:2607.13758 [nucl-ex] (2026)
- "Exploring the origin of D^0 meson elliptic flow in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV using event shape engineering," A. Gevorgyan *et al.,* arXiv:2607.11017 [nucl-ex] (2026)
- "Search for dark matter in a signature with a four-prong large-radius jet in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2607.11016 [hep-ex] (2026)
- "Search for soft unclustered energy patterns in proton-proton collisions at $\sqrt{s} = 13$ TeV using data scouting," A. Gevorgyan *et al.,* arXiv:2607.09621 [hep-ex] (2026)
- "Search for nonresonant triple Higgs boson production in the final state with six bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2607.05145 [hep-ex] (2026)
- "Study of ZZ and ZH production in the $bb\tau\tau$ final state and search for high-mass spin-0 and spin-1 resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2607.01589 [hep-ex] (2026)
- "Search for physics beyond the standard model in four and three top quark production events using proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2607.30862 [hep-ex] (2026)
- "Precision luminosity measurement in proton-proton collisions at a center-of-mass energy of 13 TeV with the CMS detector at the Large Hadron Collider," A. Gevorgyan *et al.,* arXiv:2606.26832 [hep-ex] (2026)
- "Search for long-lived particles decaying into muons in proton-proton collisions at $\sqrt{s} =$ 13.6 TeV using data scouting," A. Gevorgyan *et al.,* arXiv:2606.25840 [hep-ex] (2026)
- "Observation of electroweak production of pairs of Z bosons in proton-proton collisions at 13 TeV," A. Gevorgyan *et al.,* arXiv:2606.20558 [hep-ex] (2026)
- "Analysis of the CPstructure of the Yukawa coupling between the Higgs boson and tau leptons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV," A. Gevorgyan *et al.,* arXiv:2606.03510 [hep-ex] (2026)
- "Search for single production of a vector-like B' quark decaying to a top quark and a W boson in the single-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV," K. Tauqeer *et al.,* arXiv:2606.01423 [hep-ex] (2026)

- "Search for light scalar particles produced in Higgs boson decays in exclusive final states with two muons and two hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2605.27875 [hep-ex] (2026)
- "Search for a new heavy scalar resonance decaying to a pair of Z bosons in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2605.26462 [hep-ex] (2026)
- "Search for low-mass resonances decaying to $\tau\tau$ and measurement of the $\Upsilon \to \tau\tau$ decay in proton-proton collisions at $\sqrt{s} = 13.6$ TeV," A. Hayrapetyan *et al.,* arXiv:2605.25103 [hep-ex] (2026)
- "Search for Higgs boson decays into two neutral scalars with unequal masses in final states with b quarks and tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2605.19144 [hep-ex] (2026)
- "Search for new physics in triple boson production in proton-proton collisions at $\sqrt{s} = 13$ TeV using the effective field theory approach," A. Hayrapetyan *et al.,* arXiv:2605.15023 [hep-ex] (2026)
- "Constraints on anomalous Higgs boson couplings to vector bosons and fermions using the $\gamma\gamma$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2605.14245 [hep-ex] (2026)
- "First measurements of vector boson scattering in $W^{\pm}W^{\pm}$ and WZ production in all-leptonic final states at $\sqrt{s} = 13.6$ TeV," A. Hayrapetyan *et al.,* arXiv:2605.15396 [hep-ex] (2026)
- "Search for pair production of additional neutral scalars within the Inert Doublet Model in a final state with two electrons or two muons in proton-proton collisions at $\sqrt{s} = 13$ TeV and $\sqrt{s} = 13.6$ TeV," A. Gevorgyan *et al.,* arXiv:2605.13614 [hep-ex] (2026)
- "Search for single vector-like quark production in opposite-sign dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2605.13649 [hep-ex] (2026)
- "Search for a new heavy scalar resonance decaying into the Higgs boson and a new scalar particle in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2605.02848 [hep-ex] (2026)
- "Observation of the rare decay $\eta \to \mu^+\mu^-e^+e^-$," A. Hayrapetyan *et al.,* arXiv:2605.00615 [hep-ex] (2026)
- "Search for light charged Higgs bosons decaying to charm and strange quarks in $t\bar{t}$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2604.27795 [hep-ex] (2026)
- "Improved results on Higgs boson pair production in the 4b final state," A. Hayrapetyan *et al.,* arXiv:2604.27044 [hep-ex] (2026)
- "Measurement of the top quark pair production cross section in PbPb collisions at $\sqrt{s_{NN}} = 5.36$ TeV," A. Belyaev *et al., JHEP* **08,** 054 (2026)
- "Search for electroweakinos in compressed-spectrum scenarios with low-momentum isolated tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al., Phys. Rev. Lett.* **137,** 131802 (2026)
- "Simultaneous measurements of N-subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted W bosons and top quarks," A. Hayrapetyan *et al., JHEP* **09,** 258 (2026)
- "Measurement of the $Z \to \mu^+\mu^-$ angular coefficients in pp collisions at $\sqrt{s} = 13$ TeV as functions of transverse momentum and rapidity," A. Hayrapetyan *et al.,* arXiv:2604.25678 [hep-ex] (2026)
- "Search for associated production of a Higgs boson and two vector bosons via vector boson scattering at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al., Phys. Rev. Lett.* **137,** 101806 (2026)

- "Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2604.20999 [hep-ex] (2026)
- "Search for the single production of vector-like quarks decaying into a W boson and a b quark using single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2604.17564 [hep-ex] (2026)
- "Highly boosted dielectron identification in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al., Phys. Rev. D* **114,** 052004 (2026)
- "Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2604.14236 [hep-ex] (2026)
- "A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al., JHEP* **08**, 098 (2026)
- "Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al., Phys. Lett. B* **880,** 140812 (2026)
- "Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons," A. Hayrapetyan *et al.,* arXiv:2604.09809 [hep-ex] (2026)
- "Observation of nuclear suppression in coherent $\Upsilon(1S)$ photoproduction off heavy nuclei at the LHC," A. Belyaev *et al., Phys. Rev. Lett.* **137,** 142301 (2026)
- "Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al., Phys. Lett. B* **879,** 140671 (2026)
- "Evidence of $ZZ\gamma$ production and observation of $4\ell\gamma$ in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al., Phys. Rev. Lett.* **137,** 091805 (2026)
- "Search for Higgs boson pair production in the $b\bar{b}WW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s} = 13.6$ TeV," A. Gevorgyan *et al., JHEP* **08,** 019 (2026)
- "Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2603.25458 [hep-ex] (2026)
- "Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Gevorgyan *et al.,* arXiv:2603.23454 [hep-ex] (2026)
- "Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2603.21965 [hep-ex] (2026)
- "Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al., JHEP* **07,** 248 (2026)
- "Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass," A. Hayrapetyan *et al., JHEP* **07,** 054 (2026)
- "Jet peak shapes based on two-particle angular correlations in lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV," A. Belyaev *et al.,* arXiv:2603.14385 [nucl-ex] (2026)
- "Measurement of the t-channel single top quark cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV," A. Hayrapetyan *et al., JHEP* **08,** 223 (2026)
- "Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al.,* arXiv:2603.11035 [hep-ex] (2026)
- "Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at $\sqrt{s} = 13.6$ TeV," A. Hayrapetyan *et al.,* arXiv:2603.03689 [hep-ex] (2026)

- "Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at $\sqrt{s} = 13$ TeV," A. Hayrapetyan *et al., Phys. Rev. D* **114,** 032005 (2026)
- "Combination of ATLAS and CMS searches for Higgs boson pair production at $\sqrt{s} = 13$ TeV," G. Aad *et al., Phys. Rev. Lett.* **137,** 131803 (2026)
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